

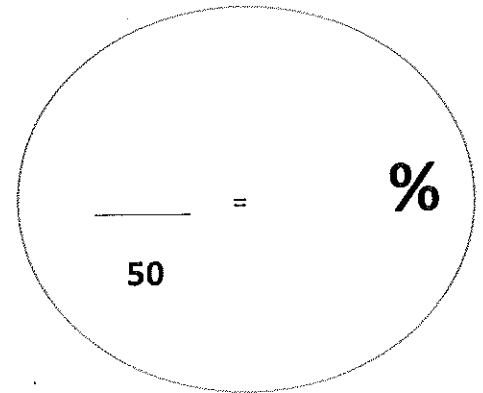


Ellenbrook
CHRISTIAN COLLEGE

Investigation 1 – Take-home
Angles and parallel lines

Name: SOLUTIONS

Time allowed: 1 week



Answer in the spaces provided.

Show all working clearly, in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. Correct answers given without supporting reasoning may not be awarded full marks.

To complete this investigation, you will need to demonstrate your ability to research well and apply knowledge learned previously in class. It is up to you to decide whether to search for answers online from reputable sources, in the Pearson textbook, using other sources, or by applying previous knowledge.

Part 1 Parallel lines

(10 marks)

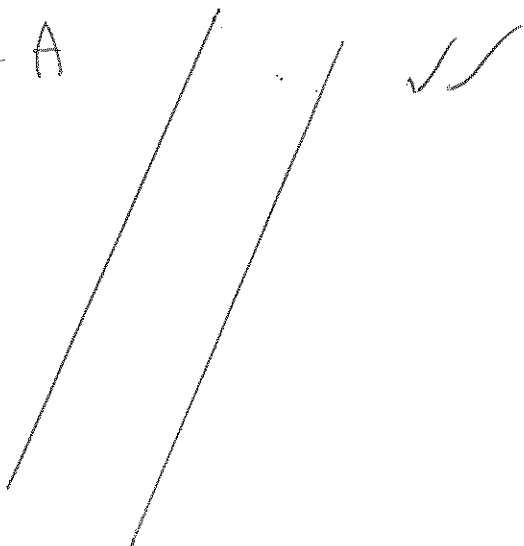
Question 1

(1,2,2 = 5 marks)

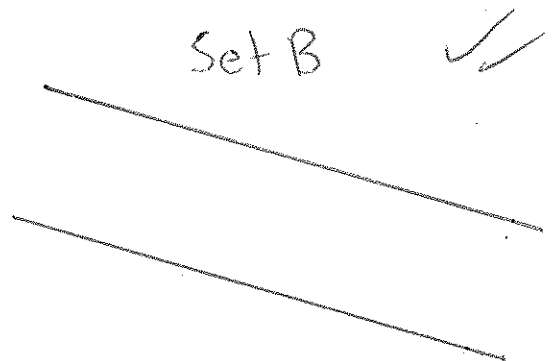
What are parallel lines? Provide a definition of parallel lines below, along with a drawing of at least 2 different sets of parallel lines

Lines that never meet ✓

Set A



Set B



Question 2

(1,2,2 = 5marks)

When graphing any lines, there are 2 important features: the gradient (slope) and the y-intercept (point where $x=0$).

a) Which of these 2 features can we compare between lines to work out if they are parallel or not?

Gradient (slope) ✓

b) If we know the gradient and y-intercept of the first line, what do we know about the line parallel to it?

It has the same gradient ✓

c) If we consider the line $y=3x+4$ (i.e. gradient of 3, y-intercept of 4), use your answers in part (a) and (b) to give either the gradient or the y-intercept of a line that is parallel to that line.

gradient = 3 ✓✓

Part 2 Corresponding angles

(10 marks)

Figure A

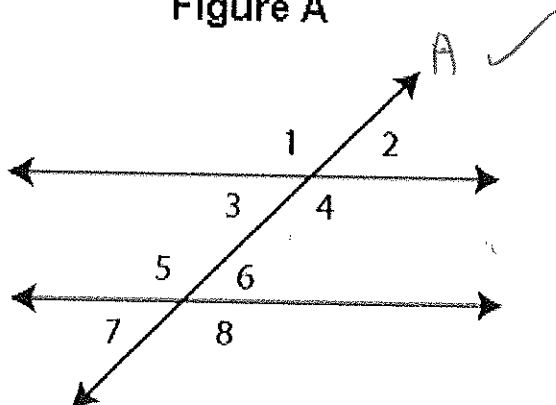
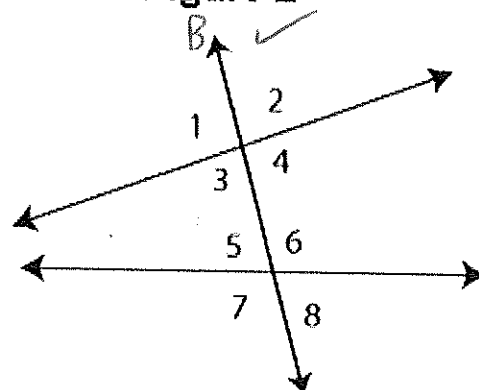


Figure B



Question 3

(2,1,1 = 4 marks)

In the next few sections, we will be looking at angles as part of lines that have been cut by a transversal. A transversal is a line which cuts at least 2 other lines. There are 3 main lines in each figure above.

a) Your task is to label the transversal line as 'A' in figure A and 'B' in figure B above.

b) Are the lines which the transversal cuts in figure A parallel or not? Why or why not?

✓ Yes, they never meet ✓ ($\frac{1}{2}$ for each)

c) Are the lines which the transversal cuts in figure B parallel or not? Why or why not?

✓ No, they will meet ✓ ($\frac{1}{2}$ for each)

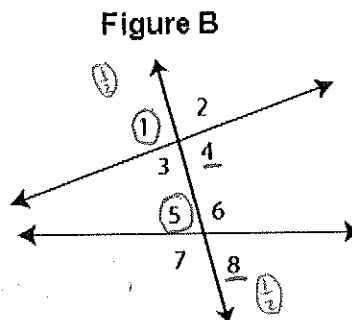
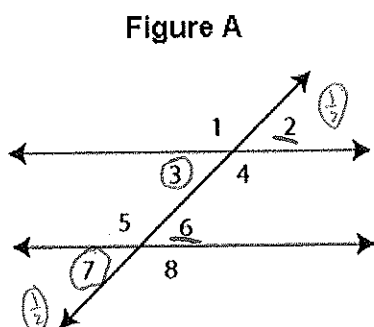
Question 4

(1,1,1,1,1,1 = 6 marks)

When considering lines cut by a transversal, there are various angles that have important properties. One of these properties relates to angles being corresponding.

a) What is the definition of a corresponding angle?

Angles in matching corners ✓



b) Circle the numbers of a pair of corresponding angles in Figure A. Underline a second pair in Figure A.

c) Circle the numbers of a pair of corresponding angles in Figure B. Underline a second pair in Figure B.

d) When a transversal cuts parallel lines, what do we know about the corresponding angles that are formed?

Corresponding angles are equal ✓

e) In figure A, are angles 1 and 8 corresponding? If they are, what is 8 if 1 is 120° ?

No ✓

f) In figure A, are angles 1 and 5 corresponding? If they are, what is 5 if 1 is 120° ?

Yes, angle 5 is equal to 1 and so is also 120° ✓

Part 3 Alternate angles

(10 marks)

Question 5

(1,1,1,1,2,2,2 = 10 marks)

Another property that is important when two lines are cut by a transversal is angles that are alternate.

a) What is the definition of an alternate angle?

Angles inside the parallel lines on opposite sides of the transversal
(ie. in Z shape) ✓

Figure A

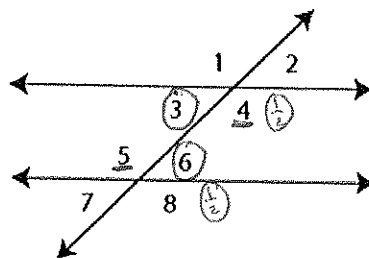
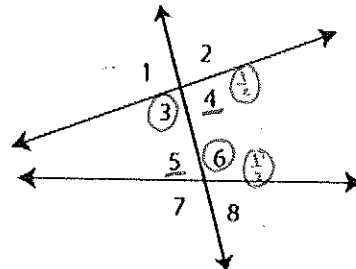


Figure B



b) Circle the numbers of a pair of alternate angles in Figure A. Underline a second pair in Figure A.

c) Circle the numbers of a pair of alternate angles in Figure B. Underline a second pair in Figure B.

d) When a transversal cuts parallel lines, what do we know about the alternate angles that are formed?

Alternate angles are equal ✓

e) In figure A, are angles 4 and 7 alternate? If they are, what is 7 if 4 is 120° ?

No ✓✓

f) In figure A, are angles 3 and 6 alternate? If they are, what is 6 if 3 is 60° ?

Yes, 6 is the same and so is 60° ✓

g) In figure B, are angles 4 and 5 alternate? If they are, is 5 if 4 is 95° ?

Yes, but lines are not parallel so we can't work it out ✓

Part 4 Co-interior angles

(10 marks)

Question 6

(1,1,1,1,2,2,2 = 10 marks)

Another property that is important when two lines are cut by a transversal is angles that are co-interior.

a) What is the definition of a co-interior angle?

Angles between the lines cut by the transversal on the same side of it.
✓

Figure A

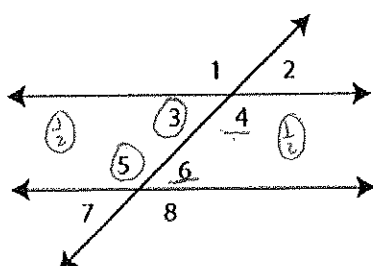
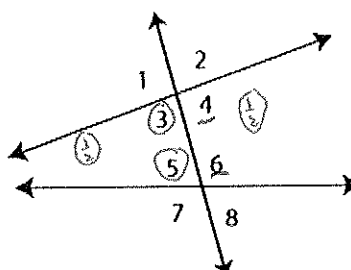


Figure B



b) Circle the numbers of a pair of co-interior angles in Figure A. Underline a second pair in Figure A.

c) Circle the numbers of a pair of co-interior angles in Figure B. Underline a second pair in Figure B.

d) When a transversal cuts parallel lines, what do we know about the co-interior angles that are formed?

Co-interior angles add up to 180° ✓

e) In figure A, are angles 3 and 5 co-interior? If they are, what is 5 if 3 is 60° ?

Yes, 5 is $180^\circ - 60^\circ = 120^\circ$ ✓

f) In figure A, are angles 3 and 6 co-interior? If they are, what is 6 if 3 is 60° ?

No ✓✓

g) In figure B, are angles 4 and 6 co-interior? If they are, what is 6 if 4 is 95° ?

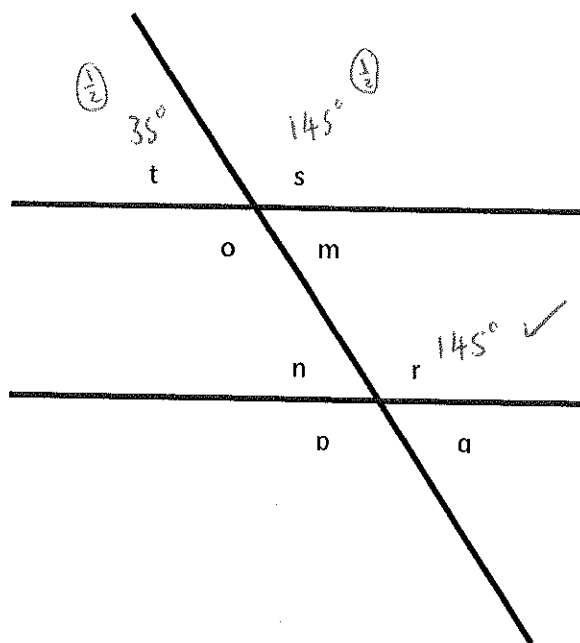
Yes, but not parallel so we can't work it out ✓

Part 5 Problem solving

(10 marks)

Question 7

(2,2,2,2,2 = 10 marks)



Extension: a potential extra 2 marks are available for those who scored less than 100% if you describe what vertically opposite angles are and what supplementary angles are, along with an example of each using the letters in this diagram.

vertically opposite:

- angles opposite each other when the 2 lines cross
- e.g. o and s (1/2)

supplementary:

- add up to 180° (1/2)
- e.g. n and r (1/2)

A set of railroad tracks is being constructed as above. The engineers only had time to measure angle m when they were constructing the tracks, but now need to provide a scale drawing of the tracks to their boss. They don't have time to go back there, as they were working a long way from home and the office. They ask you, the expert mathematician, to help avert this crisis.

Angle m was measured as exactly 35° by the engineers.

- a) The engineers need to work out angle n. What is angle n? What fact did you use to work it out?

$$n = m = 35^\circ \text{ (alternate)}$$

- b) The engineers need to work out angle o. What is angle o? What fact did you use to work it out?

$$o = 180^\circ - n = 145^\circ \text{ (co-interior)}$$

- c) The engineers need to work out angle p. What is angle p? What fact did you use to work it out?

$$p = o = 145^\circ \text{ (corresponding)}$$

- d) The engineers need to work out angle q. What is angle q? What fact did you use to work it out?

$$q = 180^\circ - p = 35^\circ \text{ (supplementary)}$$

- e) The engineers are very happy, but need you to fill in the rest for them. Write the angles of r, s and t next to their letters on the diagram above.

END OF INVESTIGATION

10/10